

KHOVD, Mongolia

WMO: 442180

Lat: 47.9958N

Lon: 91.6325E

Elev: 1405

StdP: 85.55

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.8	-32.3	-37.7	0.1	-34.4	-34.9	0.2	-31.7	9.0	-8.8	6.5	-8.4	0.9	180	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.8	15.5	28.2	15.0	26.7	14.4	18.0	25.9	16.7	25.0	15.7	23.9	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.5	13.1	20.7	13.8	11.7	19.9	12.4	10.7	19.7	57.0	26.2	52.7	25.0	49.3	24.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	7.5	5.9	-34.4	32.9	4.1	2.1	-37.3	34.4	-39.7	35.6	-42.0	36.8	-45.0	38.4
			-34.5	18.6	4.0	2.2	-37.4	20.2	-39.8	21.4	-42.0	22.6	-45.0	24.2
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-22.3	-17.3	-4.8	5.9	12.3	18.4	20.4	18.1	11.5	3.2	-7.9	-17.8
	DBStd	15.54	5.81	7.18	6.79	5.30	4.62	3.48	2.93	3.39	4.47	5.01	6.07	5.70
	HDD10.0	4046	1002	764	458	140	31	2	0	1	36	214	538	861
	HDD18.3	6219	1260	998	716	373	190	39	14	46	209	468	787	1119
	CDD10.0	1032	0	0	1	17	103	254	322	251	79	4	0	0
	CDD18.3	164	0	0	0	1	4	42	77	38	2	0	0	0
	CDH23.3	1491	0	0	0	3	55	377	658	358	39	0	0	0
	CDH26.7	336	0	0	0	0	6	77	171	78	4	0	0	0
Wind	WSAvg	1.7	0.9	1.1	1.7	2.4	2.5	2.1	1.8	1.7	1.8	1.6	1.3	1.0
Precipitation	PrecAvg	113	1	1	3	7	12	23	32	23	8	3	2	2
	PrecMax	226	8	7	15	32	51	66	119	105	29	18	11	7
	PrecMin	59	0	0	0	0	0	0	3	0	0	0	0	0
	PrecStd	35	2	2	3	6	12	17	23	20	7	4	3	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.5	3.6	14.8	22.5	27.5	31.0	33.0	31.1	26.7	19.3	12.6	4.0
		MCWB	-6.0	-1.7	6.0	9.7	13.9	14.9	18.2	15.8	13.2	9.1	4.9	-0.6
	2%	DB	-7.5	-0.2	11.6	19.5	24.7	28.7	30.3	28.8	24.2	16.1	7.9	-1.8
		MCWB	-9.9	-4.2	3.6	8.2	12.1	14.8	16.4	14.9	12.4	7.1	2.1	-5.2
	5%	DB	-11.3	-3.6	9.0	17.2	22.8	27.0	28.5	26.9	22.1	13.9	4.4	-6.1
		MCWB	-12.8	-6.7	2.0	7.2	11.2	14.2	15.5	14.6	11.3	6.0	-0.5	-8.4
	10%	DB	-14.4	-6.9	6.2	15.0	21.0	25.5	27.0	25.2	19.9	11.9	1.3	-9.6
		MCWB	-15.3	-9.0	0.4	6.0	10.0	13.6	14.9	14.0	10.4	4.7	-2.5	-11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.8	-1.4	6.5	11.2	15.3	18.4	21.2	19.0	15.4	9.9	5.4	-0.7
		MCDB	-2.4	2.9	14.4	20.8	25.6	27.4	29.9	25.1	24.1	17.7	12.6	3.7
	2%	WB	-9.9	-4.1	4.1	8.9	13.4	16.7	18.6	17.1	13.6	7.8	2.3	-5.1
		MCDB	-7.5	-0.4	10.9	17.9	22.8	25.8	25.8	25.6	21.9	14.9	7.6	-2.2
	5%	WB	-12.7	-6.7	2.3	7.6	12.0	15.5	17.3	15.9	12.1	6.4	-0.2	-8.4
		MCDB	-11.3	-3.9	8.6	16.2	21.1	24.1	25.0	24.2	20.4	13.3	4.0	-6.2
	10%	WB	-15.3	-9.0	0.3	6.4	10.7	14.6	16.2	14.9	10.8	4.9	-2.4	-11.1
		MCDB	-14.4	-6.9	5.8	14.3	19.9	23.1	24.0	22.8	19.0	11.4	1.3	-9.6
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.7	12.8	12.4	11.8	10.9	10.5	11.1	12.4	12.2	10.9	9.7
		MCDBR	12.4	13.9	14.8	14.4	14.2	13.2	12.7	13.4	14.9	14.4	13.3	12.8
		MCWBR	10.8	10.9	9.5	8.0	7.3	5.5	5.0	5.6	7.4	8.6	9.2	10.4
	5% WB	MCDBR	12.3	13.8	14.3	13.4	13.0	11.5	10.8	11.7	13.5	14.1	13.3	12.7
		MCWBR	10.7	10.9	9.4	8.1	7.6	5.9	5.5	5.8	7.5	9.0	9.4	10.3
Clear-Sky Solar Irradiance	taub	0.238	0.270	0.315	0.360	0.357	0.367	0.363	0.341	0.305	0.286	0.259	0.240	
	taud	2.350	2.276	2.228	2.199	2.268	2.327	2.371	2.405	2.456	2.432	2.364	2.315	
	Ebn at Noon	862	896	899	888	903	894	893	900	907	870	821	804	
	Edh at Noon	68	93	116	133	129	123	116	107	92	79	67	62	
All-Sky Solar Radiation	RadAvg	1.72	2.84	4.33	5.52	6.48	6.64	6.51	5.81	4.75	3.00	1.84	1.36	
	RadStd	0.09	0.12	0.22	0.24	0.42	0.38	0.37	0.21	0.20	0.21	0.12	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.34	N/A	-1.24	-1.56	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	-0.75	-1.17	N/A	N/A	N/A	N/A

Nomenclature: See separate page